

Yash Kumar Sahu

Project Associate

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Education

BENGALURU **Indian Institute of Science (IISc)**

2023 – 2024 M.Tech. Thesis

- Advised by **Prof. Pradipta Biswas** in the I3D Lab and Prof. Sadagopan Narasimhan.
- Thesis Title: Comparative Study on Image Translation GANs for Object Detection in Low-Resource Domains (presented at IEEE ICVTTS 2024).

CHENNAI **Indian Institute of Information Technology, Design and Manufacturing (IIITDM)**

2019 – 2024 B.Tech. & M.Tech. in Computer Science

- Co-founded Mars Rover club; represented university at international competitions.
- GPA: 8.44 / 10.00

Awards

ICRA 2024 **RoboMaster Sim2Real Challenge** (by Tsinghua University) YOKOHAMA, JAPAN

More Info. [↗](#) ○ Won 3rd prize globally among 30+ teams at the finals.

RoboCup 2023 **Autonomous Robot Manipulation Challenge** (by MathWorks) BORDEAUX, FRANCE

More Info. [↗](#) ○ Ranked 4th globally among 10+ countries in the finals.

ERC 2022 **European Rover Challenge** (by European Space Agency (ESA)) KRAKOW, POLAND

More Info. [↗](#) ○ Ranked 6th globally in the remote finals among 50+ teams from 10+ countries.

RoboWars 2023 **IIITDM Technical Festival** (sponsored by IEEE) CHENNAI, INDIA

More Info. [↗](#) ○ Awarded 2nd prize in the inter-university finals, competing against 12 robots.

Publications

ACM IUI 2026 Improving MR Interaction through GenAI. Yashaswi Sinha, Yash Kumar Sahu, Atharva Rege, Rubini M, Himanshu Vishwakarma, Pradipta Biswas. *International Conference on Intelligent User Interfaces*, 2026. (Under Preparation)

JRATE 2026 Gaze-Based Music Generation System for Users with Severe Speech Motor Impairment. Yash Kumar Sahu*, Taishin Aoki*, Himanshu Vishwakarma*, Pradipta Biswas. *SAGE Journal of Rehabilitation and Assistive Technologies Engineering*, 2026. (Accepted)

ICRA 2025 Blind Tactile Exploration for Surface Reconstruction. Yashaswi Sinha*, Soumojit Bhattacharya*, Yash Kumar Sahu, Pradipta Biswas. *IEEE International Conference on Robotics and Automation*, 2025.

ACM IUI 2025 Diffuse Your Data Blues: Augmenting Low-Resource Datasets via User-Assisted Diffusion. Yashaswi Sinha, Yash Kumar Sahu*, Shravan Shanmugam*, Abhishek Mukhopadhyay, Pradipta Biswas. *International Conference on Intelligent User Interfaces*, 2025.

IEEE ICVTTS 2024 Comparative Study on Image Translation GANs for Object Detection in Low-Resource Domains. Yash Kumar Sahu, Abhishek Mukhopadhyay, Gyanig Kumar, Pradipta Biswas. *International Conference on Vehicular Technology and Transportation Systems*, 2024. (Accept. Rate 30%)

IEEE CICT 2023 Vision-Based Object Sorting in Dynamic Environments using YOLO for RoboCup ARM Challenge. Yash Kumar Sahu, Radhika Mittal, Deep Patel, Chayan Maiti, M Sreekumar. *International Conference on Information Communication Technology*, 2023.

* Denotes equal contribution

Experience

- Project Associate **Intelligent Inclusive Interaction Design (I3D) Lab, IISc** Aug 2024 – Current
- Researching diffusion models, vision-language models, and 3D reconstruction for controllable image generation, assistive eye tracking, and multimodal learning, including two Government of India-funded projects (USD 150K total).
- Research Intern **Centre for AI, IoT and Robotics (CAIRO), IIITDM** Jan 2023 – Jul 2024
- Autonomous robotic manipulation for object sorting using a 7-DoF Franka Emika Panda, integrating path planning, depth estimation, and custom YOLO-based object detection and classification.
- Research Intern **Mobile Robotics Lab, IISc** May 2023 – Jul 2023
- Developed a scalable drone navigation system using corridor-based 3D path planning, layered navigation architecture, and A*, Dijkstra's, and RRT algorithms, reducing cloud memory usage by 37%.
- Research Intern **Smart Manufacturing Lab, IIITDM** May 2022 – Dec 2022
- Built a full-stack PyQt GUI for synchronized dual-arm 3D printing, including G-code-to-pose mapping and real-time visualization of robot poses and print statistics.
- Software Intern **HyperHorizon [🔗]** Aug 2022 – May 2023
- At the Undersea Systems Division, developed a ROS-based autonomous underwater vehicle (AUV) system with C++/Python navigation, a PyQt mission planner, and 3D localization using IMU, depth SONAR, and GPS sensor fusion.

Skills

- Languages & Tools Python, PyTorch, C/C++, CUDA, Bash, OpenGL, OpenMP, PyQt, PyBullet, Docker, Git, Linux, MATLAB, ROS, ROS2, NVIDIA Isaac Sim

Talks

- 2024 **Hands on Object Detection and CNNs**, Talent Sprint, IISc
- 2024 **Paper Presentation**, *IEEE ICVTTS 2024*, Amrita Vishwa Vidyapeetam, Bengaluru
- 2023 **Paper Presentation**, *IEEE CICT 2023*, IIITDM Jabalpur
- 2022 **Practical Robotics with ROS**, *4 Lecture Series*, IIITDM Kancheepuram

Volunteering

- 2021 – 2023 **Senior Under Officer, National Cadet Corps (NCC)**
- Led the university NCC wing of 400 cadets and ranked in the top 2% of the battalion.
- 2022 – 2024 **Software Team Lead, Treasurer and Student Mentor** | MaRS, IIITDM
- Autonomous rover navigation in Mars-like terrain, developing localization, mapping, and navigation algorithms for international rover competitions.
 - Co-founded the college's Mars Rover Club, earning the Pioneering Spirit Award and ISRO Satellite Centre recognition, while mentoring 30+ students and leading teams to 5th place nationally at ISRO's Rover Challenge.

Academic Referees

1. **Prof. Pradipta Biswas**, Associate Professor, Indian Institute of Science
○ Email: pradipta@iisc.ac.in
2. **Prof. Abhishek Mukhopadhyay**, Associate Professor, Amity University Kolkata
○ Email: amukhopadhyay@kol.amity.edu
3. **Prof. Sreekumar M**, Professor, IIITDM
○ Email: msk@iiitdm.ac.in